

Preparations for Intravenous Administration

Prior to infusion INTAXEL® Injection should be diluted in 0.9% Sodium Chloride Injection USP, 5% Dextrose Injection USP, 5% Dextrose and 0.9% Sodium Chloride Injection USP or 5% Dextrose in Ringer's Injection to a final concentration of 0.3 to 1.2 mg/ml. Parenteral drug products should be inspected visually for particulate matter and discoloration prior to administration whenever solution and container permit.

Upon preparation, solution may show haziness, which is attributed to the formulation vehicle. No significant loss in potency has been noted following simulated delivery of the solution through I.V. tubing containing an inline (0.22 micron) filter.

OVERDOSAGE

There is no known antidote for Paclitaxel overdose. The primary anticipated complications of overdose consist of bone marrow suppression, peripheral neurotoxicity and mucositis.

STORAGE CONDITIONS, USER INSTRUCTIONS AND PHARMACEUTICAL PRECAUTIONS

Store below 30°C. Protect from light. Solutions for infusion prepared as recommended are stable at ambient temperature (approximately 25°C) and room lighting conditions for up to 27 hours. Paclitaxel is a cytotoxic anticancer drug and, as with potentially toxic compounds, caution should be exercised in handling INTAXEL® injection. The use of gloves is recommended. If Paclitaxel solution contacts the skin, wash the skin immediately and thoroughly with soap and water. If Paclitaxel contacts mucous membranes the membranes should be flushed thoroughly with water.

Pharmaceutical Precautions

Non PVC containing administration sets such as those, which are polyethylene lined, should be used, while administering INTAXEL® injection. It should be administered through an in line filter with microporous membrane not greater than 0.22 microns.

Published data show that the presence of the extractable plasticizer DEHP [di- (2-ethylhexyl) phthalate] levels increase with time and concentration when dilutions are prepared in PVC containers. Consequently, the use of plasticized PVC containers and administration sets is not recommended. INTAXEL® solutions should be prepared and stored in glass, polypropylene, or polyolefin containers.

PRESENTATION

INTAXEL® Injection is available as 30 mg/5 ml, 100 mg/16.7 ml, 260 mg/43.4ml and 300 mg/50 ml multidose vials individually packed in a carton.

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Date of release: 04/2013

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For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory.

Paclitaxel for Injection Concentrate

INTAXEL®
INJECTION

DESCRIPTION

Paclitaxel, a parenteral antineoplastic agent is a diterpenoid taxane derivative initially obtained from the needles and branches of the Himalayan yew tree (*Taxus baccata*). The drug is presently obtained via a semi-synthetic process from *Taxus baccata*. It represents first agent from novel class of taxoids.

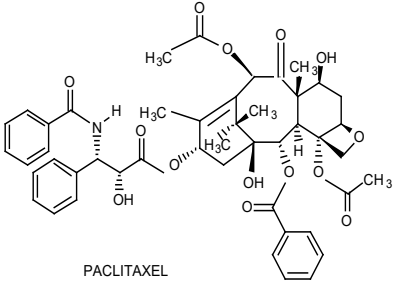
Paclitaxel for Injection Concentrate (INTAXEL® INJECTION) is a clear, almost colorless viscous solution. It is supplied as a non-aqueous solution intended for dilution with a suitable parenteral fluid prior to intravenous infusion. INTAXEL® is available in 5 ml (30 mg), 16.7 ml (100 mg), 43.4 ml (260 mg) and 50 ml (300 mg) multidose vials.

COMPOSITION

Each ml of sterile solution contains:

Paclitaxel	6 mg
Polyoxyl 35 Castor Oil	527 mg
Dehydrated Alcohol	49.7% (v/v)

CHEMICAL STRUCTURE



Paclitaxel is a white to off- white crystalline powder. Chemically it is 5B, 20-Epoxy-1, 2a, 4,7B, 10B, 13a- hexahydroxytax-11-en-9-one 4, 10-diacetate 2-benzoate 13-ester with (2R, 3 S)- N-benzoyl-3- phenylisoserine. It is highly lipophilic, insoluble in water, and melts at around 216-217°C. It has an empirical formula of C₄₇H₅₁NO₁₄ and a molecular weight of 853.9.

PHARMACOLOGY

MECHANISM OF ACTION^{1,2}

Paclitaxel exerts its effects on microtubules and their protein subunits, the tubulin dimers. Microtubules serve as facilitators of intracellular transport and maintain the integrity and function of cells. It promotes microtubule assembly by enhancing the action of tubulin dimers, stabilizing existing microtubules resulting in the inhibition of the normal dynamic reorganization of the microtubular network that is essential for vital interphase and mitotic cellular functions. In addition, paclitaxel induces abnormal arrays or "bundles" of microtubules throughout the cell cycle and multiple asters of microtubules during mitosis, resulting in the breakage of chromosomes. The G2 - and M-phases of the cell cycle are affected.

PHARMACOKINETICS^{3,4}

Paclitaxel given by IV administration exhibits a biphasic decline. The initial decline results from peripheral distribution and elimination and the second decline follows distribution from peripheral tissues to plasma. Paclitaxel does not appear to cross the blood-brain barrier to an appreciable amount; however, it does distribute to ascitic fluid. Ascitic fluid concentrations can be 40% more than serum concentrations. It gets 89-98% bound to serum proteins. In vitro studies with human liver microsomes and tissue slices showed that paclitaxel gets metabolized primarily to 6a- hydroxypaclitaxel by the cytochrome P450 isozyme CYP2C8; and to two minor metabolites, 3'- p-hydroxypaclitaxel and 6a, 3'-p-dihydroxypaclitaxel, by CYP3A4. Hepatic metabolism and biliary excretion are likely routes of elimination, given that only 10% of paclitaxel is eliminated unchanged in the urine, and high concentrations of the drug have been found in bile. Nonlinear clearance has been observed. The elimination half-life ranges from 1.3 -8.6 hours without dose dependency. After intravenous administration of 15-275 mg/m² doses of paclitaxel as 1-, 6-, or 24-hour infusions, mean values for cumulative urinary recovery of unchanged drug ranges from 1.3 to 12.6% of the dose, indicating extensive non-renal clearance.

DRUG INTERACTION⁵

- Patients receiving concomitant treatment with drugs (imidazole derived antifungal agents) affecting the cytochrome P-450 isoenzymes, CYP2C8 and CYP3A4 are to be closely monitored as the metabolism of paclitaxel is catalyzed by CYP2C8 and CAY3A4 isoenzymes.
- Administration of cisplatin prior to paclitaxel infusion, results in a 33% decrease in clearance of paclitaxel and a more significant and severe incidence of myelosuppression as cisplatin decreases paclitaxel clearance.
- Patients receiving drugs known to cause bradycardia (β-blockers, calcium channel blockers, digoxin) should be carefully monitored.
- Live virus vaccines should be administered only when clearly needed during paclitaxel therapy due to potentiation of virus replication, adverse reactions to the virus and the immunocompromised status of the patient.
- Plasma levels of doxorubicin may get increased when paclitaxel and doxorubicin are used in combination.

CLINICAL STUDIES

Ovarian cancer⁶

Combination of cisplatin-paclitaxel has produced a superior clinical response and prolonged progression free survival. Paclitaxel-cisplatin (TP) regimen was compared with cisplatin-cyclophosphamide (CP) regimen-the previous standard regimen for first line epithelial ovarian cancer in 668 women with stage IIB, IIC, III or IV disease. There was a 59% and 45% response in TP and CP group, respectively and complete response (CR) in 41% and 27% of TP and CP, respectively. Median overall survival was 35.6 months with TP compared with 25.8 months with CP. Median progression free survival was 15.5 months for TP and 11.5 months for CP. It was concluded that the combination of paclitaxel and cisplatin conferred a survival advantage over the combination of cyclophosphamide and cisplatin.

Breast Cancer^{7,8}

Paclitaxel was highly active and well tolerated in 38 patients with metastatic breast cancer in a phase II study. Patients received paclitaxel 250 mg/m² as a continuous 24-hour infusion every 21 days for a median of 6 cycles, with granulocyte-colony stimulating factor support days 3 through 10. Objective responses occurred in 62% of patients with complete responses in 3 patients (12%) and partial responses



INTAXEL®
INJECTION

For the use only of a Registered Medical Practitioner or a Hospital or a Laboratory
Paclitaxel for Injection Concentrate

Mfd. In India by:
Fresenius Kabi Oncology Ltd.
Village - Kishanpura, P.O. Guru Majra,
Tehsil - Nalagarh, Distt. Solan
(H.P.) - 174101

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in 13 (50%) An overall response rate of 51.4% (7 complete, 11 partial responses) was observed in a phase II study evaluating the combination of paclitaxel and epirubicin as second-line therapy in 35 patients with metastatic breast cancer. Grade III or IV neutropenia occurred in 30.2% of cycles initially but was decreased to 18.1% with subsequent dosage reduction. No hematopoietic growth factor therapy was administered. Other toxicities were generally mild and included thrombocytopenia, anemia, peripheral neuropathy, myalgia, nausea and vomiting, and mucositis. Cardiac toxicity was not observed at median and maximum cumulative epirubicin doses of 285 mg/m² and 630 mg/m² respectively. The first 10 patients received 6 weekly doses of epirubicin 35 mg/m² and paclitaxel 80 mg/m², repeated after a 2-week interval until response or disease progression was observed. Eight patients required a paclitaxel dosage reduction to 70 mg/m² due to neutropenia. The next 25 patients received epirubicin 25 mg/m² and paclitaxel 80 mg/m² which, again reduced to 70 mg/m² in 6 patients.

AIDS related Kaposi’s sarcoma⁹

Paclitaxel is safe and effective for advanced and previously treated Kaposi’s sarcoma associated with the acquired immune deficiency syndrome. In a phase II trial, 56 mostly male patients (aged 27 to 58 years) received paclitaxel 100 mg/m² over 3 hours every 2 weeks until complete remission, disease progression, or intolerable toxicity. A median of 10 cycles was administered. Granulocyte-colony stimulating factor treatment was initiated in 56% of patients prior to study entry. Thirty-three patients had previously received at least 1 anthracycline-containing regimen; 31 of these were considered resistant or intolerant to anthracyclines. An overall objective response was documented in 59% of patients, including 2 complete and 57 partial responses. Response rates were not different in the subgroup of patients who had previously received an anthracycline. The overall median duration of response was approximately 10 months. Facial swelling, plantar lesions and tumor-related edema improved in 72%, 91%, and 79% of patients, respectively. Grade 3 or 4 adverse effects included neutropenia (61%), anemia (27%), fatigue (25%), and myalgia (16%). Neuropathy was mostly mild to moderate (47%).

Lung Carcinoma¹⁰

The combination of escalating-dose paclitaxel (150 to 250 mg/m² infused over three hours) plus carboplatin (AUC = 6, Calvert method) produced an overall response rate of 62% among 42 patients with non-small cell lung cancer. The maximum tolerated dose of paclitaxel was 250 mg/m² and grade III neuropathy was the dose-limiting toxicity. The optimum regimen for follow-up trials was paclitaxel 225 mg/m² combined with carboplatin (AUC = 6) every three weeks.

INDICATIONS

- First-line therapy for the treatment of ovarian cancer in combination with cisplatin
- Treatment of breast cancer after failure of combination chemotherapy for metastatic disease or relapse within 6 months of adjuvant chemotherapy. Prior therapy should have included an anthracycline unless clinically contraindicated
- As second-line treatment of AIDS-related Kaposi’s sarcoma
- In combination with other anticancer drugs to treat common forms of advanced lung cancer

CONTRAINDICATIONS

- Patients with a known hypersensitivity to paclitaxel or other drugs formulated in Cremophor® EL (polyoxyethylated castor oil)
- Paclitaxel should not be used in patients with solid tumors who have baseline neutrophil counts of <1500cells/mm³ or in patients with AIDS-related Kaposi’s sarcoma with baseline neutrophil counts of <1000cells/mm³
- It is also contraindicated during pregnancy and lactation

ADVERSE EFFECTS^{11,12,13}

Haematologic

Bone marrow suppression is the major dose-limiting toxicity of paclitaxel. Neutropenia, the most important hematologic toxicity, is dose and schedule dependent and is generally rapidly reversible. Neutropenia does not appear to increase with cumulative exposure and is neither more frequent nor more severe for patients previously treated with radiation therapy. Use of supportive therapy, including G-CSF, is recommended in case of severe neutropenia. 12% of all treatment courses report fever. Thrombocytopenia is uncommon. Anemia (Hb <11 g/dl) is observed in 78% of patients and is severe (Hb <8 g/dl) in 16% of the cases. No consistent relationship between dose or schedule and the frequency of anemia is observed.

Hypersensitivity Reactions (HSR)

The frequency and severity of HSRs are not affected by the dose or schedule of paclitaxel administration. These are observed in 20% of all courses and in 41% of all patients. The most frequent symptoms observed are dyspnea, flushing, chest pain and tachycardia. Rare reports of chills and back pain in association with HSRs have been received.

Cardiovascular

Hypotension, during the first 3 hours of infusion, occurs in 12% of all patients and 3% of all courses administered. The frequency of hypotension and bradycardia are neither influenced by prior anthracycline therapy, nor by the dose or the schedule. Significant cardiovascular events possibly related to paclitaxel occur in approximately 1% of all patients. These events include syncope, rhythm abnormalities, hypertension and venous thrombosis. The arrhythmias include asymptomatic ventricular tachycardia, bigeminy and complete AV block requiring pacemaker placement. ECG abnormalities are noted in 23% of the patients. Congestive heart failure has been reported typically in patients who have received other chemotherapy, notably anthracyclines.

Central Nervous System

The frequency and severity of neurologic manifestations are dose-dependent, but are not influenced by infusion duration. Peripheral neuropathy is observed in 60% of all patients (3% severe) and in 52% (2% severe) of the patients without pre-existing neuropathy. The frequency increases with cumulative dose. Neurologic symptoms are observed in 27% of the patients after the first course of treatment and in 34-51% from course 2 to 10. Sensory symptoms usually improve or resolve within several months of paclitaxel discontinuation. The incidence of neurologic symptoms does not increase in the subset of patients previously treated with cisplatin. Pre-existing neuropathies resulting from prior therapies are not a contraindication for paclitaxel therapy. Other serious neurologic events have been rare (<1%) and include grand mal seizures, syncope, ataxia and neuroencephalopathy.

Gastrointestinal

Mild to moderate nausea/ vomiting, diarrhea and mucositis are reported by 52%, 38% and 31% of all patients, respectively. Mucositis is schedule dependent and occurs more frequently with the 24-hour than with the 3-hour infusion. Rare reports of intestinal obstruction, intestinal perforation, pancreatitis, ischemic colitis, and dehydration have been received.

Kidney / Genitourinary

Urinary tract infections are frequently reported infectious complications. Among the patients treated for Kaposi’s sarcoma with paclitaxel, renal toxicity of grade III or IV severity has been reported.

Liver

No relationship is observed between liver function abnormalities and either dose or schedule of paclitaxel administration. Among patients with normal baseline liver function 7%, 22% and 19% had elevations in bilirubin, alkaline phosphatase and AST (SGOT), respectively. Prolonged exposure to paclitaxel is not associated with cumulative hepatic toxicity. Rare reports of hepatic necrosis and hepatic encephalopathy leading to death have been documented.

Respiratory

Upper respiratory tract infections do occur. Rare reports of interstitial pneumonia, lung fibrosis and pulmonary embolism have been received. Rare reports of radiation pneumonitis have been received in patients receiving concurrent radiotherapy.

Musculo-skeletal

There is no consistent relationship between dose and schedule of paclitaxel and the frequency or severity of arthralgia/ myalgia. 60% of all patients treated experience arthralgia/myalgia; 8% experience severe symptoms. The symptoms are usually transient, occur 2 - 3 days after paclitaxel administration, and resolve within a few days. The frequency and severity of musculoskeletal symptoms remain unchanged throughout the treatment period.

Injection site reactions

These include reactions secondary to extravasation, are usually mild and consist of erythema, tenderness, skin discoloration, or swelling at the injection site. These are observed more frequently with the 24-hour infusion than with the 3-hour infusion. Rare reports of more severe events such as phlebitis, cellulitis, induration, skin exfoliation, necrosis and fibrosis have been documented. It is advisable to closely monitor the infusion site for possible infiltration during drug administration.

Other Clinical Events: Alopecia is observed in almost all (87%) of the patients. Nail changes are uncommon (2%). Edema has been reported in 21% of all patients. Rare reports of skin abnormalities related to radiation recall as well as reports of maculopapular rash and pruritus have been documented.

PRECAUTIONS AND WARNING^{11,12,13}

- Contact of the undiluted concentrate with plasticized polyvinyl chloride (PVC) equipment or devices used to prepare solutions for infusions is not recommended. In order to minimize patient exposure to the plasticizer DEHP [di- (2-ethylhexyl) phthalate], which may be leached from PVC infusion bags or sets, diluted INTAXEL® solutions should preferably be stored in bottles (glass, polypropylene) or plastic bags (polypropylene, polyolefin) and administered through polyethylene lined administration sets.
- INTAXEL® should be administered through an in line filter with a microporous membrane not greater than 0.22 microns.
- Paclitaxel (INTAXEL®) should be administered under the supervision of a physician experienced in the use of cancer chemotherapeutic agents. Appropriate management of complications is possible only whenadequate diagnostic and treatment facilities are readily available.
- All the patients should be pretreated with corticosteroids (such as dexamethasone), diphenhydramine or promethazine hydrochloride and H₂ antagonists (such as cimetidine or ranitidine) before receiving INTAXEL®. Severe hypersensitivity reactions characterized by dyspnea and hypotension requiring treatment, angioedema, and generalized urticaria have been reported in 2% of the patients receiving Paclitaxel. These reactions were probably histamine mediated.
- Bone marrow suppression (primarily neutropenia) is dose dependent and is the dose limiting toxicity. Neutrophil nadirs have been reported at a median of 11 days. INTAXEL® should not be administered to patients with baseline neutrophil counts of less than 1500 cells/mm³. Frequent monitoring of blood counts should be instituted during INTAXEL® treatment. Patients should not be retreated with subsequent cycles of INTAXEL® until neutrophils recover to a level of >1,500cells/ mm³ and platelets recover to a level of >100,000cells/mm³.
- Severe conduction abnormalities have been documented in less than 1% patients during INTAXEL® therapy. In one case insertion of a pacemaker was required due to the recurrence of atrioventricular block. In another case, Paclitaxel infusion was completed despite the occurrence of asymptomatic recurrent atrioventricular block and a pacemaker was subsequently placed to allow further Paclitaxel treatment. If patients develop significant conduction abnormalities during INTAXEL® administration, appropriate therapy should be administered and continuous cardiac monitoring should be performed during subsequent therapy with INTAXEL®.
- Paclitaxel has been shown to be clastogenic *in vitro* (chromosomal aberrations in human lymphocytes) and in vivo (micronucleus test in mice). Paclitaxel was not mutagenic in the Ames test of CHO/HGPRT gene mutation assay. The carcinogenic potential of paclitaxel has not been studied.
- Administration of paclitaxel prior to and during mating produced impairment of fertility in male and female rats at doses equal or greater than 1 mg/kg/day. At this dose, paclitaxel has caused reduced fertility and reproductive indices, and increased embryo- and fetotoxicity.

Use in pregnancy and lactation

Paclitaxel may cause foetal harm when administered to a pregnant woman. It has been shown to be embryo and foetotoxic in rats and rabbits and to decrease fertility in rats. In these studies, Paclitaxel was shown to result in abortions, decreased corpora lutea, a decrease in implantations and live foetuses, and increased resorptions and embryo-foetal deaths. No gross external, soft tissue or skeletal alterations occurred. There are no studies in pregnant women. If INTAXEL® is used during pregnancy, or if the patient becomes pregnant while receiving this drug, the patient should be apprised of the potential hazard. Women of childbearing potential should be advised to avoid becoming pregnant during therapy with INTAXEL®.

It is not known whether paclitaxel is excreted in human milk. Following intravenous administration of carbon-14 labeled paclitaxel to rats on days 9 to 10 postpartum, concentrations of radioactivity in milk were higher in milk than in plasma and declined in parallel with the plasma concentrations. Because many drugs are excreted in human milk and because of the potential for serious adverse reactions in nursing infants, it is recommended that nursing be discontinued when receiving paclitaxel therapy.

DOSAGE AND ADMINISTRATION¹⁴

All patients should be premedicated prior to INTAXEL® administration in order to prevent severe hypersensitivity reactions.

The recommended doses of INTAXEL® is 175 mg/m² body surface area by intravenous infusion over 3 hrs and repeated after 3 weeks.

Course of INTAXEL® should not be repeated until the neutrophil count is at least 1,500 cells/mm³ and the platelet count is at least 1,00,000 cells/mm³. Patients who experience severe neutropenia (neutrophils < 500 cells/mm³ for a week or longer) or severe peripheral neuropathy during INTAXEL® therapy should have dosage reduced by 20% for subsequent courses of INTAXEL®. The incidence and severity of neurotoxicity and haematological toxicity increases with dose, especially above 190 mg/m².

Premedication regimen with Paclitaxel

All the patients should be pretreated with steroids, antihistamines and if needed H₂ -receptor antagonist as given below:-

Dexamethasone 8 mg I.V. or 20 mg orally 12 and 6 hours before therapy and

Promethazine Hydrochloride injection 25 mg, 30 minutes before therapy and

Cimetidine 300 mg I.V. or Ranitidine 50 mg I.V. 30 minutes before therapy.

Preparation and Administration Precautions

Paclitaxel is a cytotoxic anticancer drug and like other potentially toxic compounds, caution should be exercised in handling INTAXEL® Injection. The use of gloves is recommended. If INTAXEL® solution contacts the skin, wash the skin immediately and thoroughly with soap and water. If INTAXEL® contacts the mucous membranes, the membranes should be flushed thoroughly with water.

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